

QUESTION 2 :

Simple School Attendance Tracker

Your school wants to track the attendance of students for one week.

Task:

1. Create a **list of student names** (minimum 3 students).
2. Create a **2D list** to store attendance, where each inner list contains **5 boolean values** (true for present, false for absent) representing attendance for 5 days.
3. For each student:
 - Count the number of days they were present.
 - Calculate their attendance percentage using the formula:

$$\text{Attendance \%} = \frac{\text{Number of Present Days}}{\text{Total Days}} \times 100$$

4. Display the student's name along with their attendance percentage, rounded to **two decimal places**.

PROGRAM:

```
void main() {  
  
    // List of student names
```

```
List<String> students = ["veeresh", "manju", "sham"];

int totaldays=5;

// each inner list = 5 days of attendance (true = present, false =
absent)

List<List<bool>> attendance = [

    [true, true, false, true, true],

    [true, false, true, true, true],

    [true, true, true, false, true],

];

const int totalDays = 5;

print("Attendance Report:");

print("total days tracked:$totaldays");

for (int i = 0; i < students.length; i++) {

    int daysPresent = attendance[i].where((present) => present).length;

    double attendancePercent = (daysPresent / totalDays) * 100;

    attendancePercent = double.parse(attendancePercent.toStringAsFixed(2));

    print("${students[i]}: $attendancePercent%");

}

}
```

OUTPUT:

```
Attendance Report:  
total days tracked:5  
veeresh: 80.0%  
manju: 80.0%  
sham: 80.0%
```

QUESTION 3 :

Grocery Store Discount Calculator (12 marks)

A grocery store offers discounts based on total purchase value.

Task:

Write a Dart program to:

1. Store grocery items in a `Map<String, double>` (item $\rightarrow$ price)
2. Allow adding at least **5 items** with their prices.
3. Calculate the total bill and apply:
 - 5% discount if total < 500
 - 10% discount if total ≥ 500 and < 1000
 - 15% discount if total ≥ 1000
4. Display:
 - All items with prices

- Total before discount
- Discount amount
- Final bill after discount

PROGRAM:

```
void main() {
    Map<String, double> groceryItems = {
        "Milk": 120.50,
        "Bread": 80.00,
        "Eggs": 150.00,
        "Rice": 350.00,
        "Apples": 200.00,
    };

    double totalBeforeDiscount = groceryItems.values.reduce((a, b) => a + b);

    double discountRate;
    if (totalBeforeDiscount < 500) {
        discountRate = 0.05;
    } else if (totalBeforeDiscount < 1000) {
        discountRate = 0.10;
    } else {
        discountRate = 0.15;
    }

    double discountAmount = totalBeforeDiscount * discountRate;
    double finalBill = totalBeforeDiscount - discountAmount;

    totalBeforeDiscount =
double.parse(totalBeforeDiscount.toStringAsFixed(2));
    discountAmount = double.parse(discountAmount.toStringAsFixed(2));
    finalBill = double.parse(finalBill.toStringAsFixed(2));
    print("Items Purchased:");
    groceryItems.forEach((item, price) {
        print("- $item: \${price.toStringAsFixed(2)}");
    });
}
```

```
print("\nTotal before discount:
\\${totalBeforeDiscount.toStringAsFixed(2)}");
print("Discount amount: \\${discountAmount.toStringAsFixed(2)}");
print("Final bill after discount: \\${finalBill.toStringAsFixed(2)}");
}
```

OUTPUT:

```
Items Purchased:
- Milk: $120.50
- Bread: $80.00
- Eggs: $150.00
- Rice: $350.00
- Apples: $200.00

Total before discount: $900.50
Discount amount: $90.05
Final bill after discount: $810.45
```